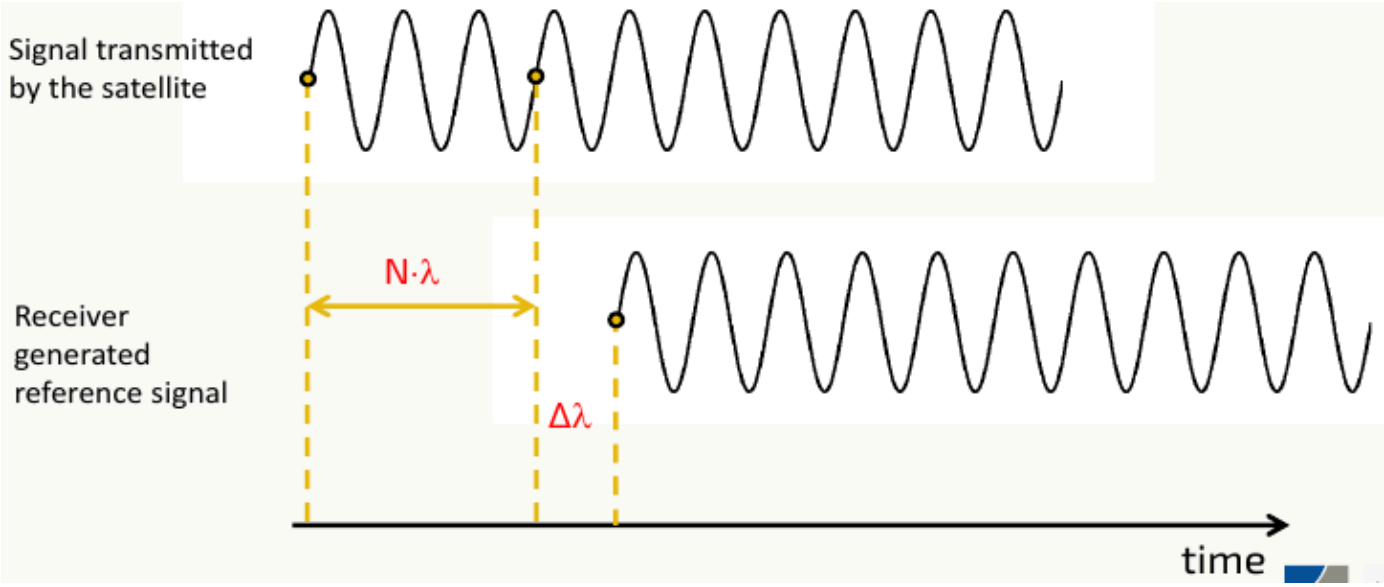
Lecture 4: Observation Errors

Carrier Phase Measurement

Correlation of generated and incoming wave after removing the code and the data from the signal

- Observation: phase difference $\Delta\lambda$ (fraction of a full wavelength)
- Problem: Ambiguities N (Number of full cycle between transmitter and receiver) have to be solved



Code-Measurement (Pseudorange):

- τ Signal transit time(ca. 70-90ms)
- t true GPS time of signal reception
- $t - \tau$ true GPS time of signal transmission
- $t^S(t - \tau)$ emission time stamp (satellite clock)
- $t_u(t)$ arrival time stamp (receiver clock)

$$p(t) = c[t_u(t) - t^S(t - \tau)]$$

Pseudorange [m]

- u ... user
- s ... satellite
- c ... speed of light

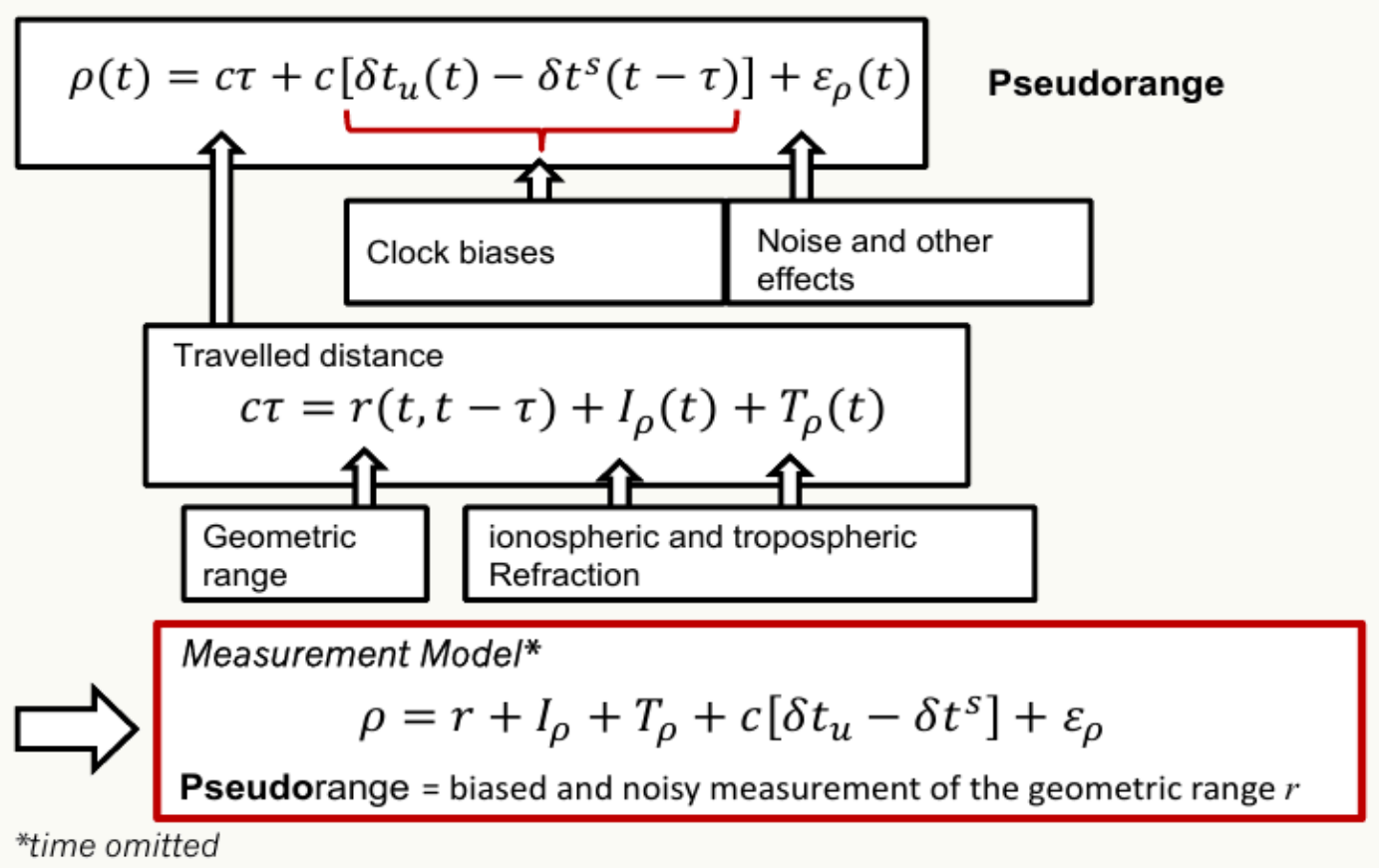
Receiver clock bias
$t_u(t) = t + \delta t_u(t)$
Unknown and time varying

Satellite clock bias

$t^S(t-\tau) = (t-\tau) + \delta t^S(t-\tau)$

Monitored by the control segment and sent via navigation message

$$
\begin{aligned}
p(t) &= c[t_u(t) - t^S(t-\tau)] + \varepsilon_p(t) \\
&= c[t + \delta t_u(t) - (t-\tau) + \delta t^S(t-\tau))] + \varepsilon_p(t) \\
&= c\tau + c[\delta t_u(t) - \delta t^S(t-\tau)] + \varepsilon_p(t)
\end{aligned}
$$



Pseudorange Measurement

$$\rho(t) = c[t_u(t) - t^s(t-\tau)]$$

Carrier Phase Measurement

Signal transit time

$$\Phi(t) = \Phi_u(t) - \Phi^s(t-\tau) + N \quad \text{[cycles]}$$

Receiver generated reference signal

Satellite transmitted received signal

Unknown number of full cycles

CARRIER PHASE MEASUREMENT

$$\Phi(t) = \Phi_u(t) - \Phi^s(t-\tau) + N$$

phase distance time

$$\Phi = \frac{1}{\lambda} \cdot r = \frac{c}{\lambda} t$$

wavelength

ambiguities are time independant!

$$\varphi = 1/\lambda\,[r + I_\varphi + T_\varphi] + c/\lambda\,[\delta t_u - \delta t^s] + N + \varepsilon_\varphi$$

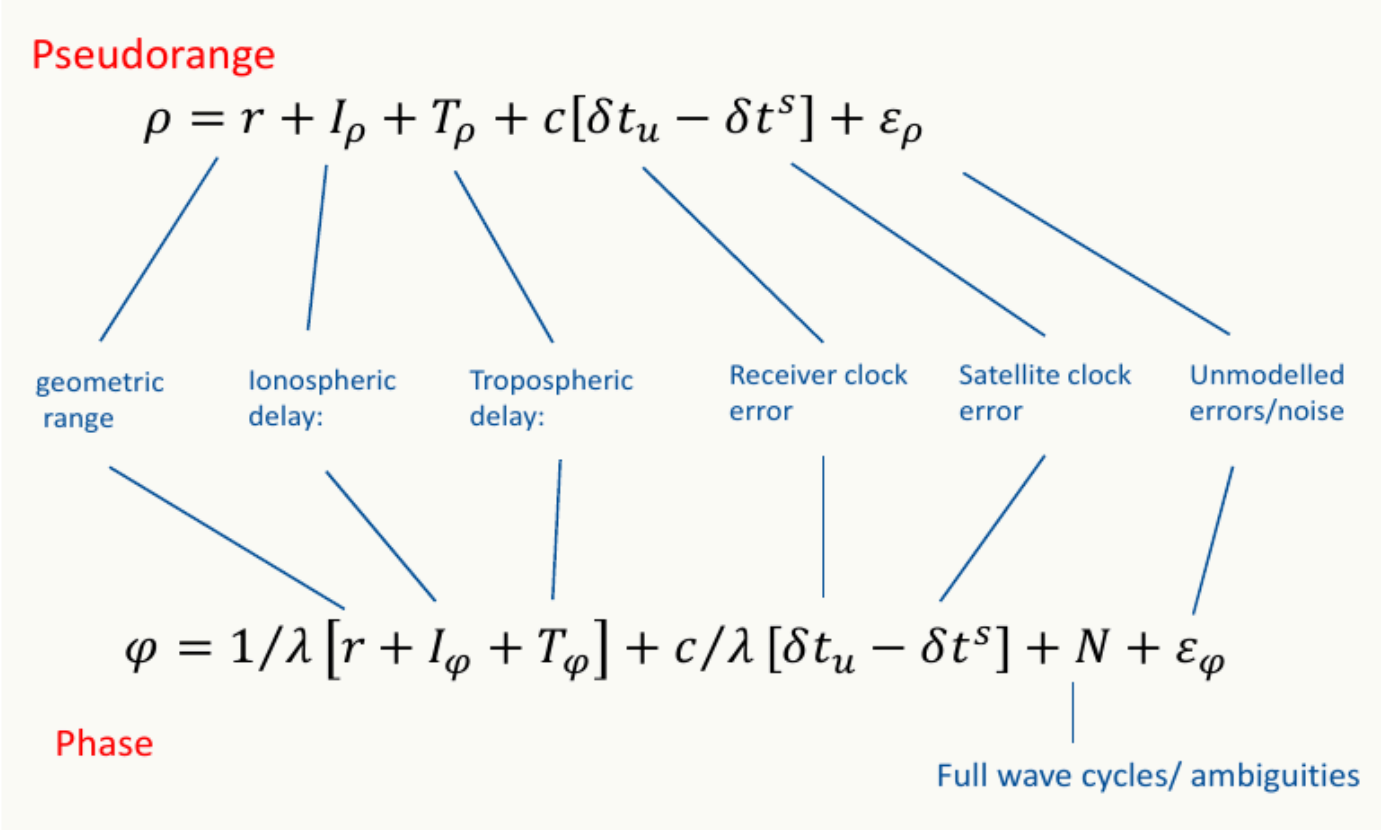
Geometric range

ionospheric and tropospheric Refraction

Clock biases

noise + other effects

**time omitted*



Observation errors

1 - Satellite Errors

- Satellite clock bias
- Satellite orbit errors
- Other satellite effects

Orbit Errors

- Ephemeris errors: radial, along track, cross track
- Radial errors has the biggest influence, BUT is the smallest

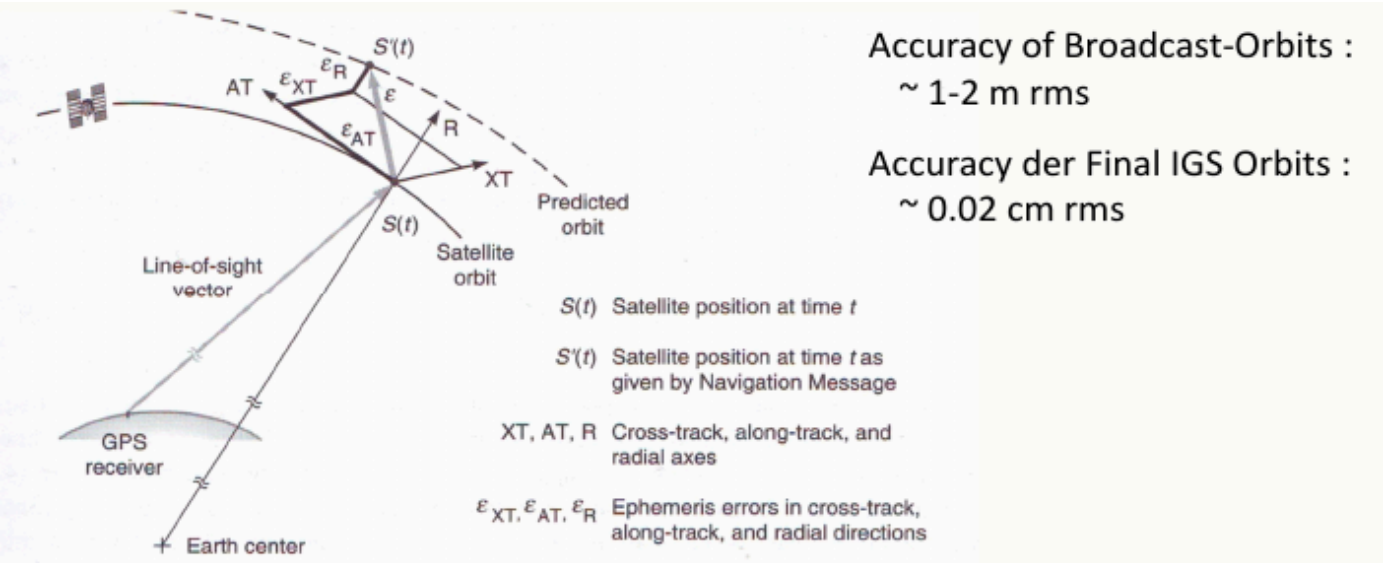


Abb.: Satellitenbahnabweichung, Quelle: Misra und Enge (2001), Fig. 4.4

Clock Errors

$$\delta t^s = t^s - t = a_{f_0} + a_{f_1}(t - t_0) + a_{f_2}(t - t_0)^2 + \Delta t_r$$

t_0	Reference epoche
a_{f0}	clock offset [s]
a_{f1}	Frequency offset [s/s]
a_{f2}	Frequency drift [s/s^2]
Δt_r	relativistic Correction

Transmitted via Navigation Message

Accuracy of corrected satellite clock: ~ 5-10 ns rms

=> 1.5 – 3 m

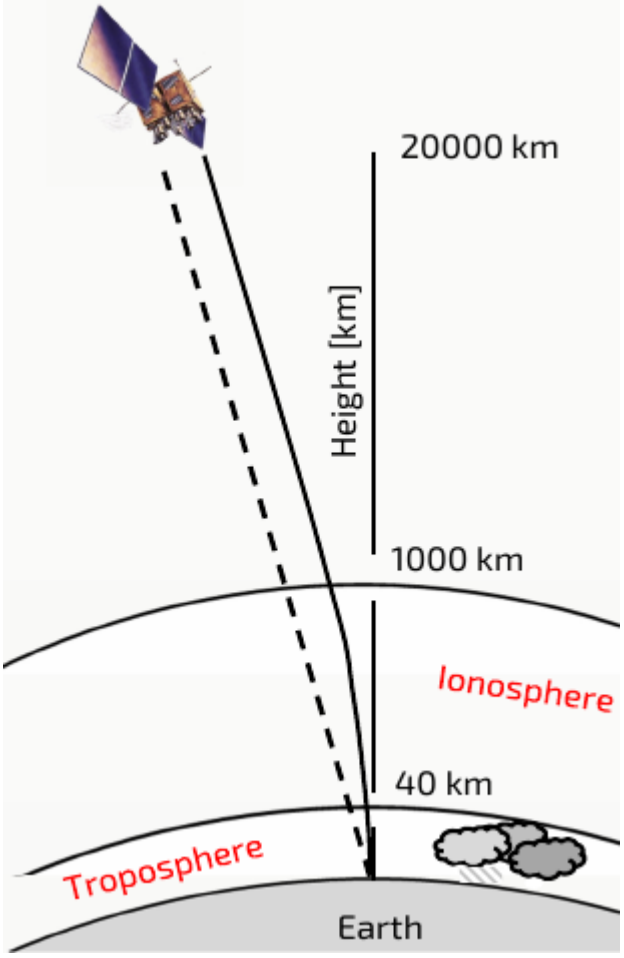
IGS Satellite clock corrections:
~ 75 ps rms

=> 0.02 m

2 - Signal Path Errors

- Ionospheric Refraction
- Troposheric Refraction
- Multipath

Influence of the Atmosphere



- Above 1000km: Vakuum -> Signal propagation @ speed of light c
- between ~50 and 1000km: Ionosphere
- below 40 km: Troposphere
- Atmosphere = Ionosphere + Troposphere
- The atmosphere changes the velocity (speed + direction) of the GPS signals!
- -> Atmospheric Refraction

Properties of a medium:

Refractive index n of a medium :

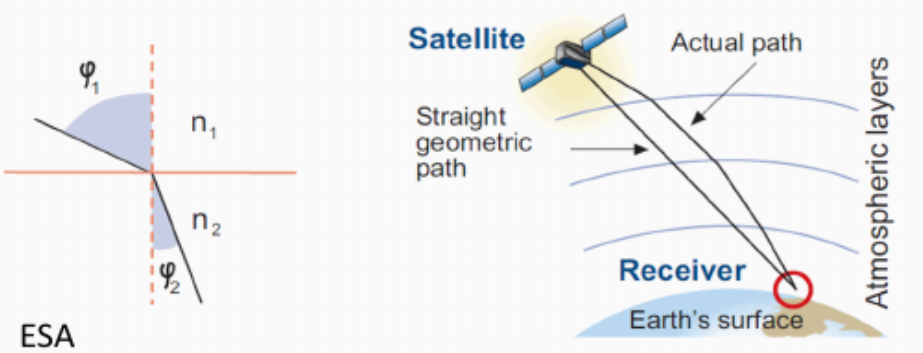
$$n = c/v$$

c ... speed of light in vacuum

v ... speed of light in the medium

A change of the refractive index in a medium induces a **bending of the signal**.

Snell's Law $n_1 \cdot \sin\varphi_1 = n_2 \cdot \sin\varphi_2$



Additional path length:

Troposphere:

~ 0cm @ elevation > 20°
~ 2cm @ elevation ~10°
~17cm@ elevation ~ 5°

Ionosphere:

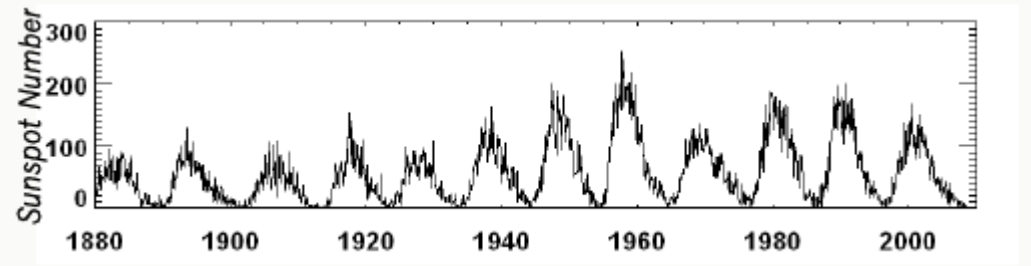
up to a few cm for low elevation

Change in range measurement

$$\Delta p = \int_S^R [n(l) - 1] dl [m]$$

-> a change of the refractive index along a path n(l) also indices a change in transit time τ

Ionospheric Delay

- Ionosphere consists of ionised gas (free electrons and ions)
- Ionisation is caused by suns' radiation
- Time varying variation
  - More free electrons and ions with sunrise
  - Electron density increases until midday, then decrease
  - No ionisation during the night
  - daily variations due to different sun activities and magnetic effects
  - Seasonal sun activities
  - 11 year cycles
  - 

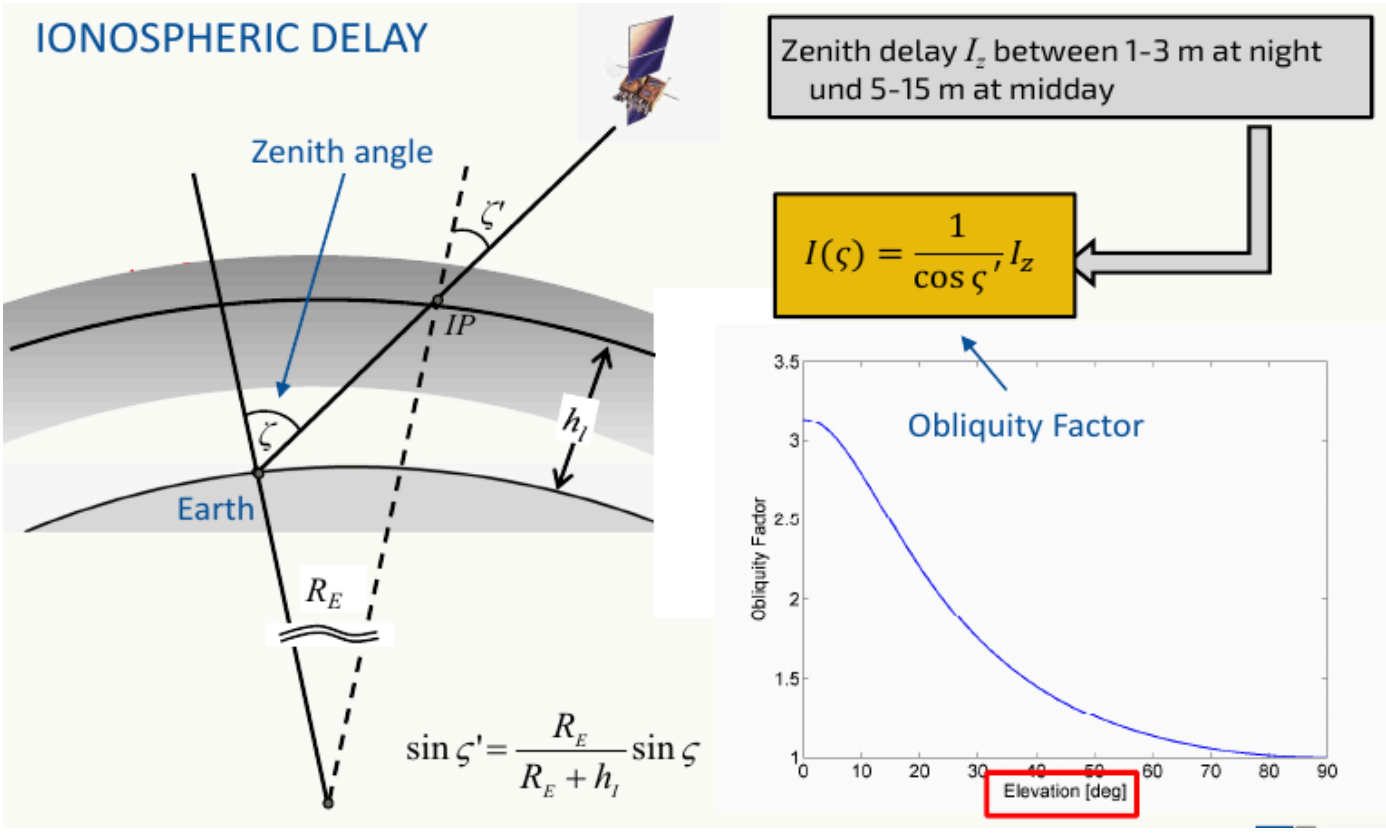
- Information about ionospheric delay is given by the Total Electron Content (TEC): # free electrons within a $1m \times 1m$ column along the signal path
- 1 Tecu = 10^{16} Electrons/m^2
- TEC maps are provided by the IGS ("Space Weather")
- TEC maps are usually given in **zenit direction** (VTEC, shortes path)
- Typical TEC values are 0.150TECU with 20-25% variations
- 1 TECU -> ~ 16cm delay @GPS L1

Ionosphere is a Dispersive Medium

- Dispersive Medium: refractive index n(f) depends on the frequency of the signal
- For GNSS (L-Band) signal: **ionosphere is a dispersive medium**, the troposphere is not
- Propagation speed is different for different frequencies
- There are phase (carrier) and group (code) velocities v_p and v_g

Ionospheric Delay:

- Phase advance
- Group delay
- Ionospheric Delay for code and for phase measurements have a different sign, but same size



Ionospheric Delay Summary

- Carrier and its modulated signal (code) propagate at different speeds through the ionosphere
  - -> **Code** measurements are **too long**, **Carrier phase** measurements are **too short**

- Phase advance and code delay are **proportional to the TEC**
- Length of the signal path through the ionosphere depends on the **Elevation of the satellite**
- **Stronger** influence for **lower** satellites
- Multiplicator: obliquity factor

Tropospheric Delay

- **Dry** (gases N_2, O_2 and **wet** part (water vapor)
- Troposphere is **not dispersive** all frequencies have same speed
- $N = N_d + N_w \rightarrow T = 10^{-6} \int_S^R N(l)dl = ... = T_d + T_w$ [m]
- Delays depend on air temperature and air pressure and air humidity
- Dry and wet part have differnet influence (dry: 90%, wet: 10%)
- Dry delay: 2-10m (high to low elevation), easy to model, low variability
- Wet delay: up to 40cm, difficult to model, high variability (weather)

Correction

1. Modeling the Zenith delay $T_Z = T_{z,d} + T_{z,w}$
2. Applying a multiplicator (oblique factor) to account for different signal length path through the troposphere (similar to ionosphere)

$$T(E) = T_{z,d} \cdot m_d(E) + T_{z,w} \cdot m_w(E)$$

Elevation

Mapping functions

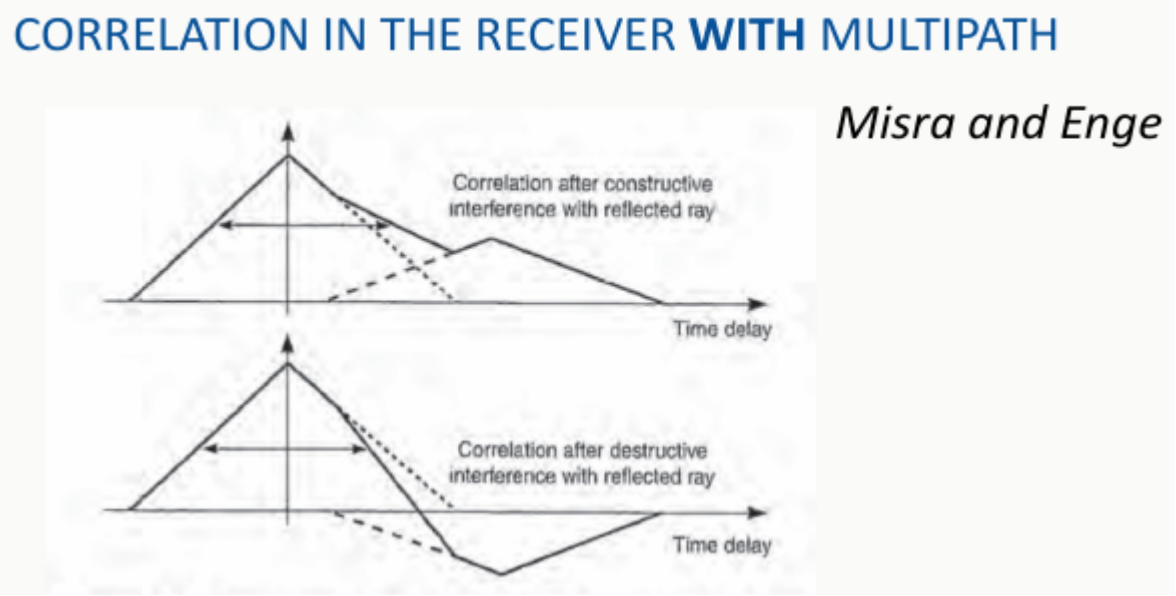
	Total	Dry Part	Wet Part
Delay	2.3 m (Zenith) 25 m (5° El.)	90%	10%, max. 40 cm
Modelling		easy	difficult
Temporal variations		slow	fast
Spatial variations		slow	fast

		Ionosphere	Troposphere
Variability		High: daily, seasonal, Solar cykles (11y), Solar flares	small
Zenith delay		Meter up to tens of meter	2.3 – 2.6 m (at sea level)
	30°	1.8	2
Obliquity Factor	15°	2.5	4
	3°	3	10
Modelling error		1-10 m	5-10 cm (without Meteorological data)
Dispersive for GNSS freq		yes	no

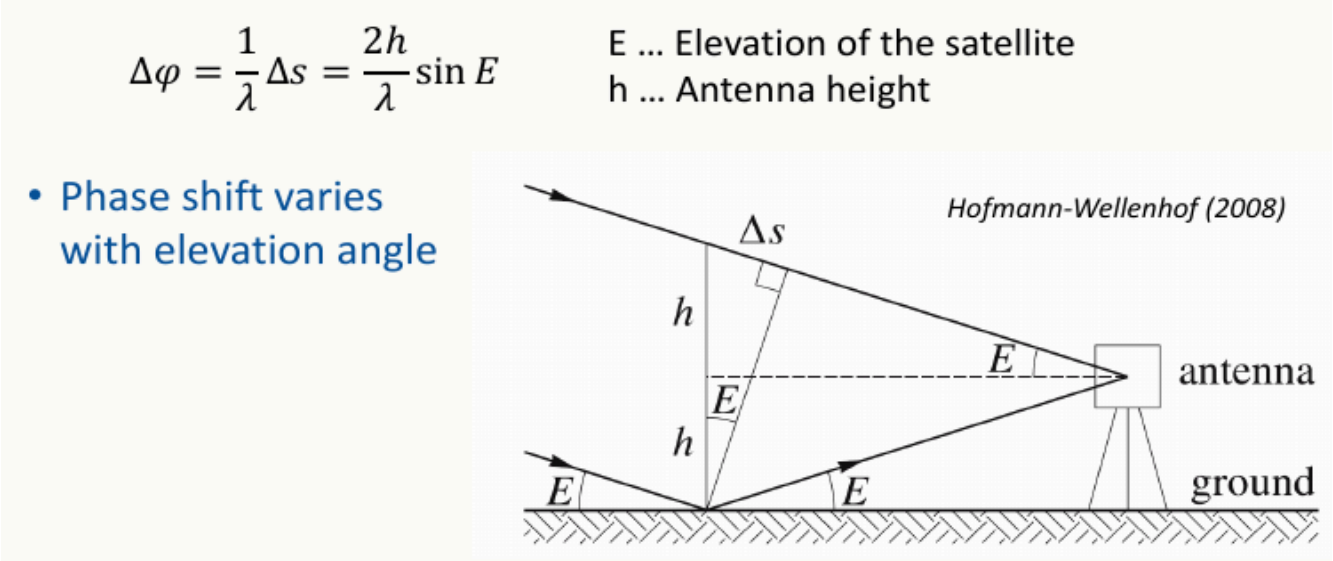
3 - Multipath

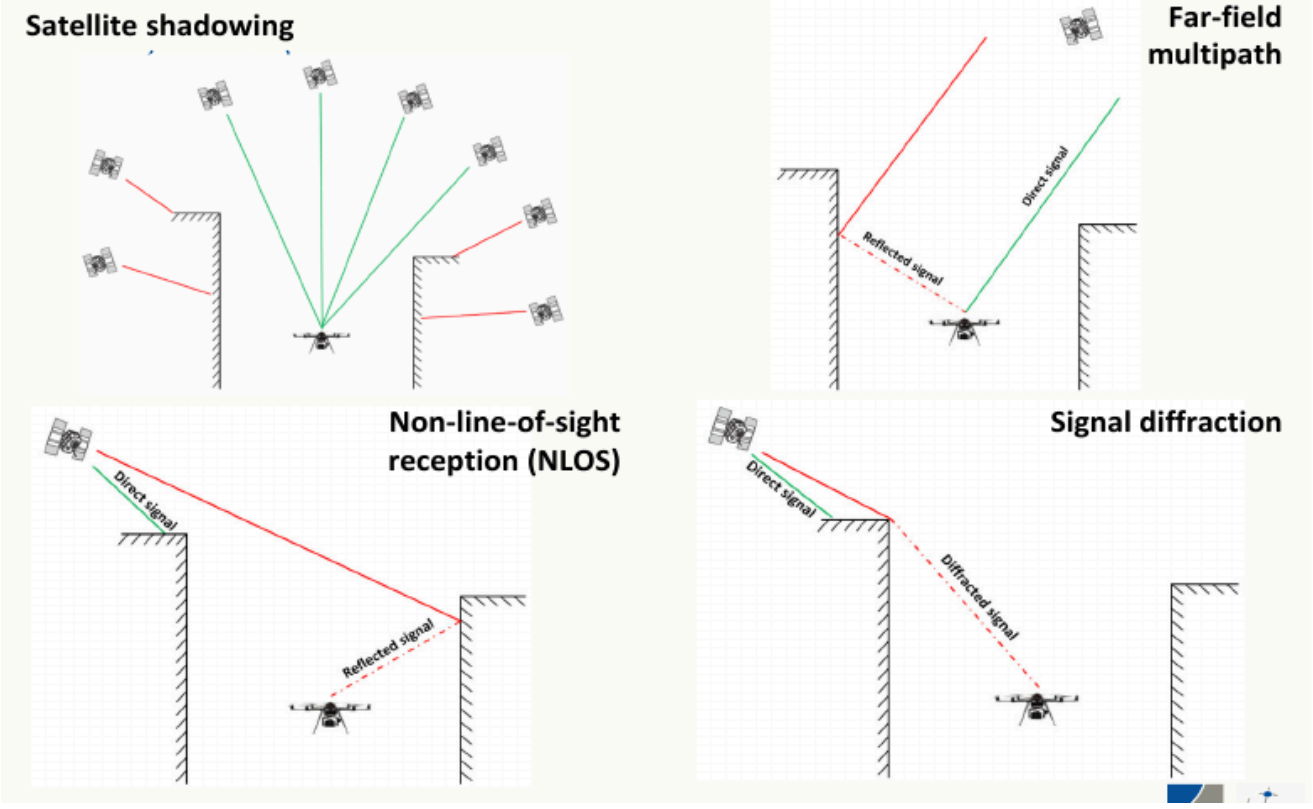
(And other antenna environment related errors)

- Arrival of signal via **more than one path**, due to reflective surfaces
- Low elevation satellites are more prone to multipath reflections
- Multipath affects code and phase measurements
- Destructive and constructive interference of direct and reflected signals



- Theoretical maximum value for code observation is $1.5T_{Chip}$ (450 m for C/A code). However, values > 15 are rarely observed
- Theoretical maximum value for phase observations is ¼ λ (5cm for GPS L1)
- Maximum possible multipath delay at ~ 5cm for carrier phases
- Multipath is periodical

- Geometry between satellites, reflectors and antenna changes with time!
- 

  - cycle time depends on the distance between the reflector to the receiver
  - The bigger the distance, the smaller the cycle time
  - Maximal cycle time is around 30-40min
- Effects can be measured out by long (> 1h) **static** measurements
- 

  Summary

  - Multipath results from reflecting surfaces in the vincinity of the antenna
  - Multipath depends on the wave length or chip length of the signal
  - Code measurements are more influenced than carrier phase measurements
    - Observation errors for **code**: **1- 15m**, theoretically up to 450 m
    - Observation errors for **carrier phases**: up to 5cm

    - Influence is period: max. cycle time usually around **30-40min** (depending on the distance of the reflecting surface to the antenna)
    - Non-line of sight reception (which is **not** multipath) can be much **more critical** than multipath

Avoidance

- Avoid reflecting surfaces around the antenna
- Do a static measurment with more than 1h observation time (multipaht is averaged out due to its periodicity)
- Use advanced antenna techniques (e.g. choke-ring antennas)

Antenna

Electromagnetic waves have polarisation perpendicular to direction of motion (Right-Handed Circular Polarisation)

- Suppression of signals from **below the horizon**
- Reflection of a signal leads to **change in polarisation** (RHCP->LHCP)
- -> GNSS antennas try to suppress LHCP signals

High precision measurements

- **Phase Center Offset**: Vector between Phase centre (Reference for GNSS signals) and ARP (Antenna Reference point)
    - -> PCO ~cm
- **Phase center variations**
    - different for different elevation and azimuth
    - PCV: Phase Centre Variation (azimuth elevation depending correction to the PCO)
    - -> PCV -> ~mm

-> For high precision measurements: Antenna Calibration (Determination of PCO and PCV)
- other wise errors ~cm

Antenna Types

- Microstrip/Patch Antennas
    - Common / mass market
    - electrical conducting patch or strip on dielectric material
- Helix Antennas
- Turnstile, Pinwheel, Spiral
- Choke ring antennas
    - Concentric rings with $\lambda/4$ depth around antenna element

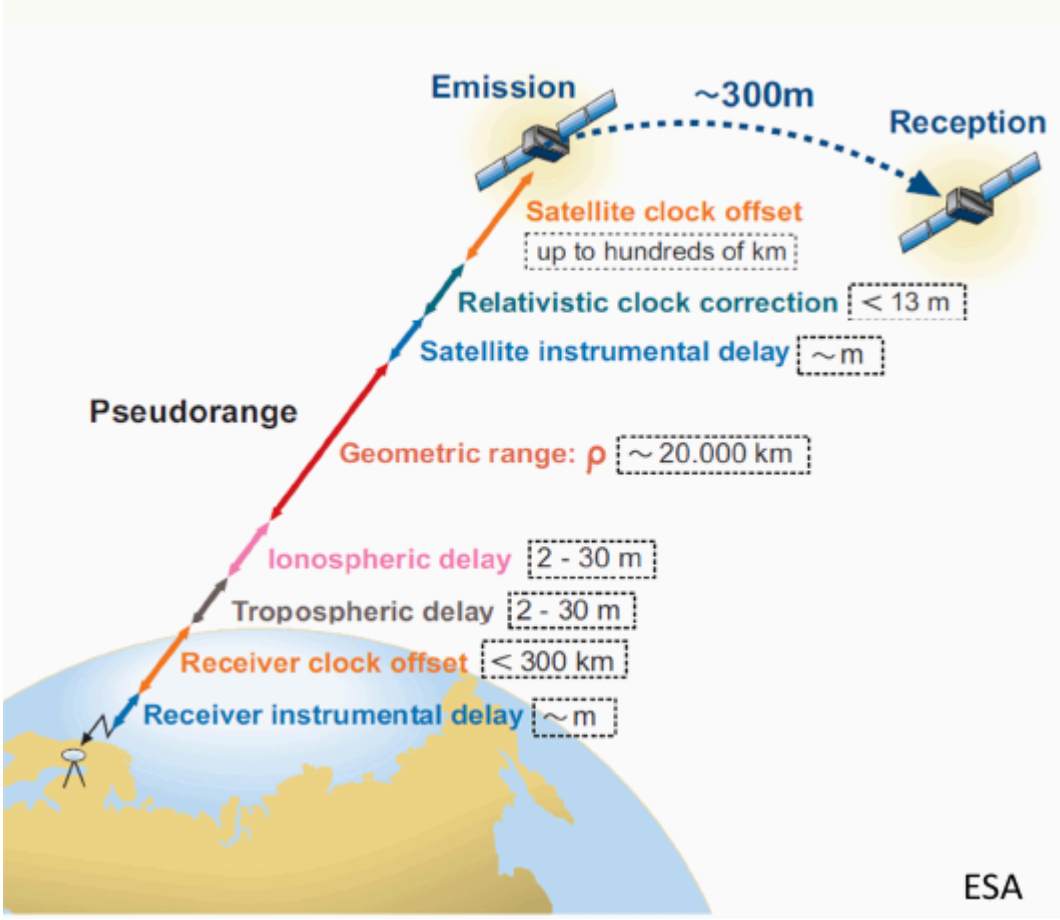
    - primary and secondary waves are eliminated
    - direct signals are not affected

Summary

- Use of correct antenna calibration parameters is important
- Relative GNSS: same antennas at both stations
    - Antenna errors eliminated in differencing process
- High accuracy: individual calibrated antennae with large ground plane (e.g. Choke Ring)
- In field:
    - correct orientation towards north for azimuth calibration parameters
    - Accurate antenna height
    - Antenna type and serial number

Receiver

- Receiver noise is the **only true random error** in GNSS
- Error: 1% of a cycle length
- Receiver clock error
    - estimated during position calculation
    - Eliminated for relativ/differential GNSS
- Receiver Environment: objects < 50cm
    - Tripod, Cable, other equipment
- Other errors:
    - Spoofing: imitation of GNSS signal
    - ->wrong position estimate
    - Jamming: Disturb and interfere original GNSS signal (1 Watt Jammer already enough)
    - Controlled Antenna Gain pattern antennas against jamming


Emission
~300m
Reception
Satellite clock offset
up to hundreds of km
Relativistic clock correction
< 13 m
Satellite instrumental delay
~m
Pseudorange
Geometric range: ρ
~20.000 km
Ionospheric delay
2 - 30 m
Tropospheric delay
2 - 30 m
Receiver clock offset
< 300 km
Receiver instrumental delay
~m
ESA